Univerzita Komenského v Bratislave
Fakulta matematiky, fyziky a informatiky

Automatic Citation Integrity Verification for LLM-Assisted Academic Writing

Diploma thesis

2027

Bc. Vladimír Tischler

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Študijný program:	Aplikovaná Informatika
Študijný odbor:	Informatika
Školiace pracovisko:	Katedra aplikovanej informatiky
Školiteľ:	Mgr. Marek Šuppa

Bratislava, 2027
Bc. Vladimír Tischler

50985887

Comenius University Bratislava
Faculty of Mathematics, Physics and Informatics

THESIS ASSIGNMENT

Name and Surname:	Bc. Vladimír Tischler
Study programme:	Applied Computer Science (Single degree study, master II. deg., full time form)
Field of Study:	Computer Science
Type of Thesis:	Diploma Thesis
Language of Thesis:	English
Secondary language:	Slovak

Title: Automatic Citation Integrity Verification for LLM-Assisted Academic Writing

Annotation: Large language models can produce plausible-looking bibliographies and in-text citations that may be partially or fully fabricated, contain metadata errors, or fail to support the surrounding claim. This thesis aims to design and evaluate a pipeline that extracts and normalizes reference strings and in-text citation anchors, resolves them against scholarly metadata sources to detect non-existent or inconsistent works, and assesses whether the cited work plausibly supports the local citation context (at sentence/paragraph level). The intended outcome is a practical verification component that flags high-risk citations with interpretable error categories (e.g., non-resolving reference, mismatched venue/year/authors, weak topical alignment, unsupported claim-to-citation link).

Aim: The goals of the thesis include (but are not limited to):
Curating an evaluation set of academic passages with citations, including LLM-generated references, with labels for existence, metadata correctness, and context support.
Benchmarking reference parsing and normalization approaches for noisy, style-diverse bibliographic strings, emphasizing robustness to incomplete metadata.
Developing a citation-resolution module (identifier recovery + fuzzy matching) that aims to separate "non-existent" from "real-but-misformatted/mismatched" references.
Prototyping a lightweight "citation support" check that aims to score whether the cited work aligns with the claim made near the citation, and analyzing failure modes.
Evaluating operational trade-offs (precision at high recall, latency/cost, domain shift) and proposing reporting thresholds suitable for editorial/educational use.

Literature:
Walters, W. H. "Fabrication and errors in the bibliographic citations generated by ChatGPT." Scientific Reports (2023). https://www.nature.com/articles/s41598-023-41032-5
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Supervisor:	Mgr. Marek Šuppa
Department:	FMFI.KAI - Department of Applied Informatics
Head of department:	doc. RNDr. Tatiana Jajcayová, PhD.
Assigned:	12.12.2025
Approved:	13.12.2025

prof. RNDr. Roman Ďurikovič, PhD.
Guarantor of Study Programme

Student

Supervisor

Statutory Declaration

. . . .
Vladimír Tischler

Acknowledgements

Abstract

Keywords:

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Bibliography

[1] C. Alvarez et al. Zero-shot scientific claim verification using LLMs and citation text. In *Proceedings of the 2024 ACL Workshop on Scholarly Document Processing (SDP)*. Association for Computational Linguistics, 2024.

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